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# **Real-Time Chat Moderation System Using Rule-Based NLP and Multithreading**

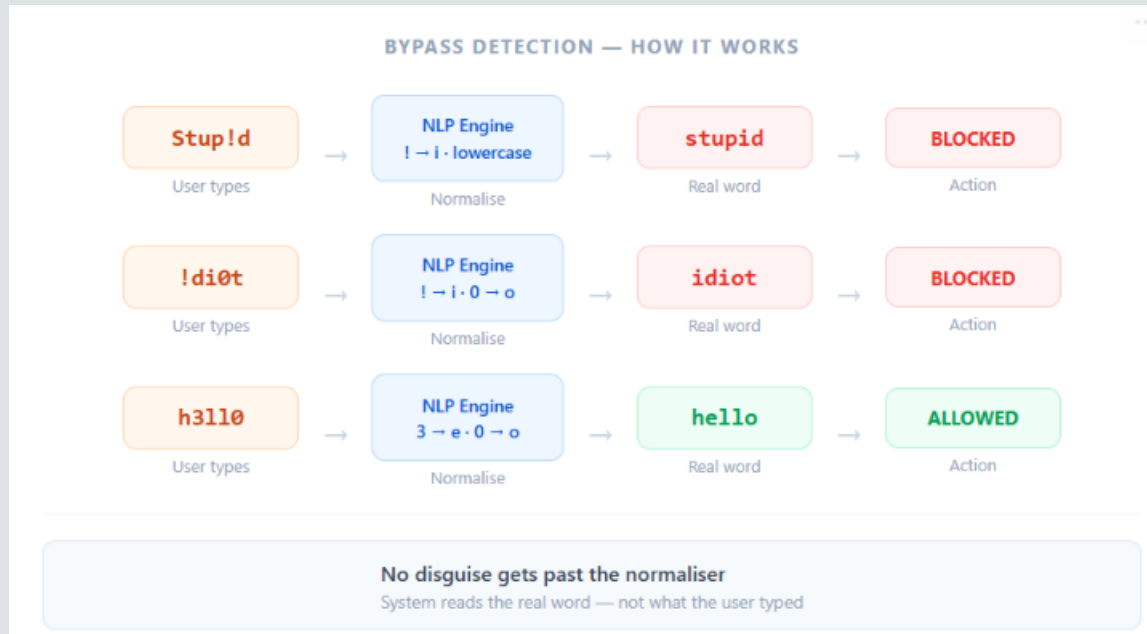
**CSA0927- Programming In Java**

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# INTRODUCTION

- Online chats move too fast for humans to monitor every single message manually.
- We are building an automated system to instantly read and filter live chat streams.
- It uses Natural Language Processing (NLP) to understand the text and spot bad words, spam, or harassment.
- It uses Multithreading to process thousands of messages at the exact same time so the chat never slows down.



- The main goal is to keep online communities safe and clean without causing any lag for the users.



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## AGENDA

- Introduction
- Problem Statement
- Objective
- Architecture Diagram
- Modules
- Conclusion
- References



# PROBLEM STATEMENT

- Popular chat applications receive massive amounts of messages every second, making human moderation impossible to scale.
- Without immediate moderation, chats quickly get ruined by spam, toxic behavior, and malicious links.
- Older, single-threaded moderation tools create bottlenecks, causing message delivery to lag during busy hours.
- Basic keyword blockers are too simple and easily tricked when users misspell bad words on purpose.

# OBJECTIVE

- To develop an automated chat moderation system capable of intercepting, analyzing, and filtering user messages in real-time.
- To implement a rule-based Natural Language Processing (NLP) engine for accurately identifying profanity, spam, and toxic text patterns.
- To utilize multithreading techniques to process high volumes of concurrent messages without introducing latency or lag to the chat interface.
- To design a dynamic decision engine that automatically passes, redacts, or blocks inappropriate messages based on strict, customizable rule sets.
- To build an intuitive frontend experience using HTML, CSS, and JavaScript where users can chat seamlessly while administrators monitor flagged content and manage rule dictionaries.

# Module 1: Real-Time Message Ingestion & Thread Manager

- Acts as the front door for the moderation system by instantly intercepting every single message sent by users.
- Maintains a continuous, open connection with the chat interface to ensure zero delay in receiving data.
- Monitors the incoming traffic volume to detect sudden spikes in chat activity during peak streaming times.
- Uses a multithreaded architecture to avoid the bottleneck of processing messages in a slow, single-file line.
- Creates and manages a pool of lightweight worker threads that are always ready to take on incoming data.
- Instantly assigns each new message to the next available worker thread for parallel processing.

## Module 2: Rule-Based NLP Processing Engine

- Functions as the analytical brain of the system, operating independently inside every individual worker thread.
- Immediately strips away unnecessary characters, extra spaces, and confusing punctuation to clean the raw text.
- Converts all incoming text to lowercase to prevent users from hiding bad words using random capital letters.
- Cross-references the cleaned message against a massive, built-in dictionary of known profanity and toxic words.
- Applies advanced regex patterns to catch complex variations of bad words, such as users replacing letters with numbers.
- Detects repetitive spam behaviors by analyzing how often identical or highly similar phrases are sent in a short time.

## Module 3: Decision & Action Controller

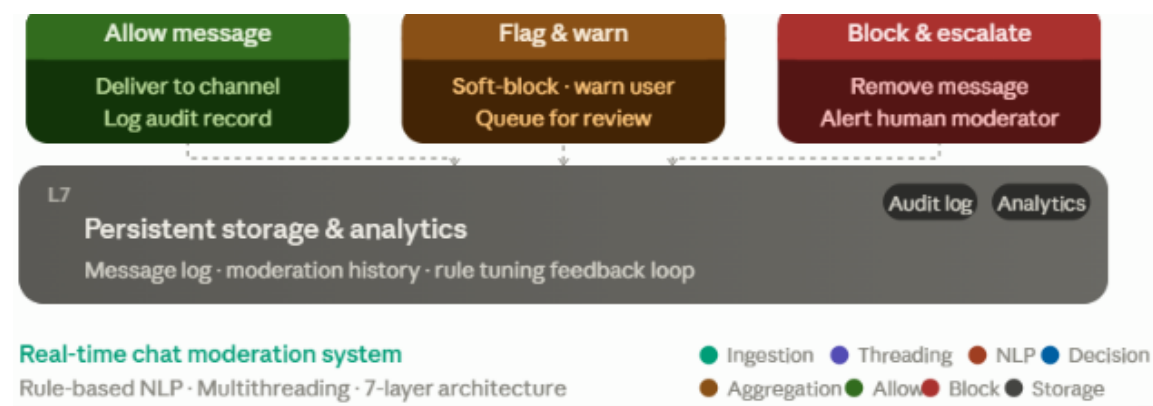
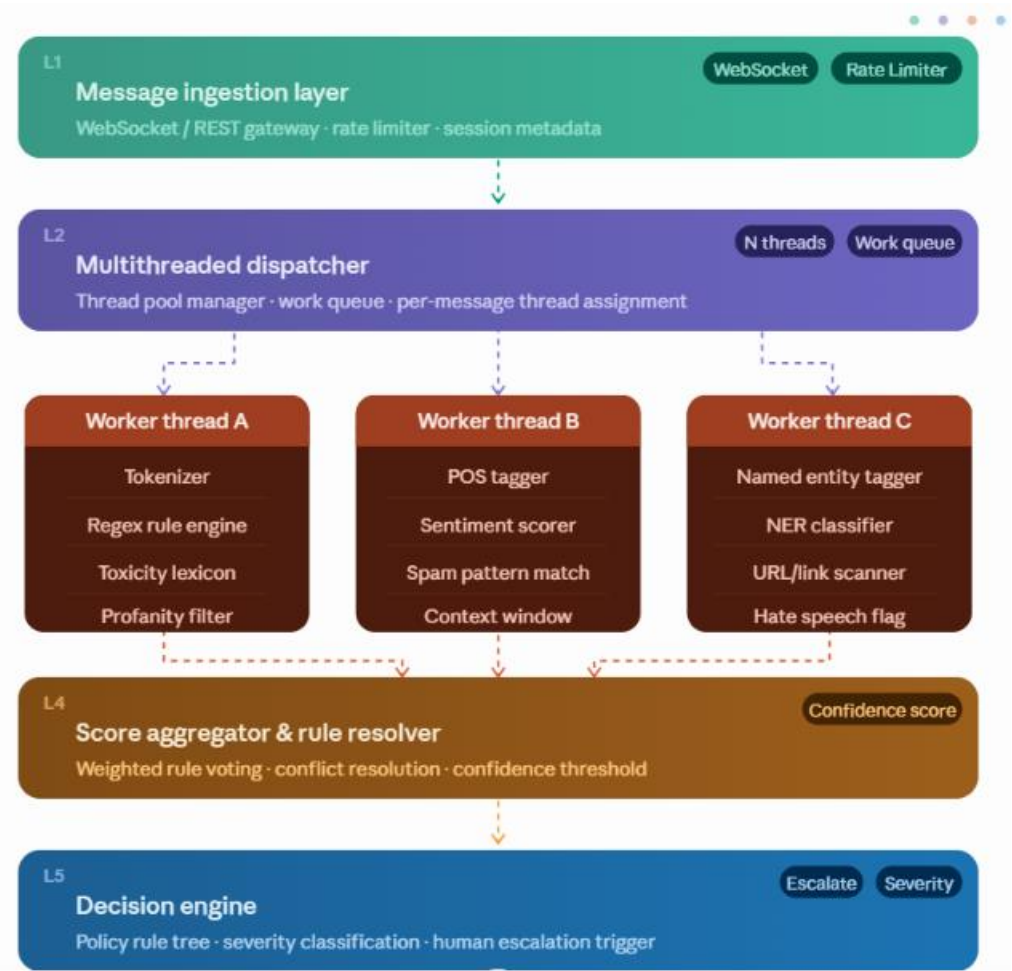
- Receives the detailed text analysis from the NLP engine to determine the final fate of the user's message.
- Calculates a specific risk score for every message based on the severity and number of the broken rules.
- Acts instantly on that risk score without requiring any human confirmation to maintain the real-time speed of the chat.
- Automatically approves completely clean messages and routes them straight to the public chat for everyone to read.
- Precisely targets and replaces isolated bad words with asterisks while keeping the rest of the safe message intact.
- Entirely blocks and deletes messages that cross the highest threshold for severe toxicity, hate speech, or dangerous links.



## Module 4: Admin Dashboard & Log Management

- Provides a secure, private web interface built exclusively for system administrators and community managers.
- Displays real-time analytics on chat health, system performance, and active thread counts in an easy-to-read layout.
- Allows moderators to instantly add new internet slang or emerging spam phrases to the system's blacklist.
- Enables admins to adjust the overall strictness level of the moderation rules without needing to restart the system.
- Silently records every single flagged violation, redacted word, and blocked message into a secure backend database.
- Creates a permanent history of user behavior so repeat offenders can be easily identified and permanently banned.

# ARCHITECTURE DIAGRAM



# Conclusion

- The project successfully proves that manual, slow chat moderation can be replaced by an efficient, fully automated system without sacrificing accuracy.
- By leveraging multithreading, the system effectively eliminates the bottleneck problem, allowing thousands of concurrent messages to be processed simultaneously without slowing down the chat for users.
- The Rule-Based NLP engine goes beyond basic keyword blocking, successfully identifying disguised profanity, bypassed words, and spam patterns that traditional filters miss.
- The immediate interception of toxic messages creates a significantly healthier online environment, protecting users from harassment and cyberbullying in real-time.
- Through the administrative dashboard, the system remains highly flexible, allowing community managers to instantly update rules and dictionaries as new internet slang and spam tactics evolve.

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*Thank  
you!*